THE DUALIZING FUNCTOR ON FINITE LENGTH MODULES OVER A LOCAL RING IS THE LINEAR DUAL INTO THE INJECTIVE HULL

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Allowed standard prerequisites (editorial).

- Matlis duality: the functor into the injective hull induces an anti-equivalence on finite length modules and is its own inverse. Invoked inside the counted proof (source lines 3220-3222 and 3227). Gives the anti-equivalence and the double-dual identity.
- The statement that an autoequivalence of the category of finite length modules is isomorphic to the identity. Invoked inside the counted proof (source lines 3223-3225). Identifies the composed functor with the identity.

Source reference. Original lemma lemma-equivalence-finite-length: sources/stacks-dualizing/pinned-original-source.tex, source lines 3173–3173. Navigation pointer only: the counted text refers to the Stacks item lemma-equivalence-finite-length, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original proposition proposition-matlis: sources/stacks-dualizing/pinned-original-source.tex, source lines 1185–1185. Navigation pointer only: the counted text refers to the Stacks item proposition-matlis, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local ring with normalized dualizing complex $\omega_A^\bullet$. Let E be an injective hull of κ. Then there exists a functorial isomorphism*

$$R\operatorname{Hom}_A(N, \omega_A^\bullet) = \operatorname{Hom}_A(N, E)[0]$$

for N running through the finite length A-modules.

Proof. By induction on the length of N we see that $R\operatorname{Hom}_A(N, \omega_A^\bullet)$ is a module of finite length sitting in degree 0. Thus $R\operatorname{Hom}_A(-, \omega_A^\bullet)$ induces an anti-equivalence on the category of finite length modules. Since the same is true for $\operatorname{Hom}_A(-, E)$ by Proposition proposition-matlis we see that

$$N \longmapsto \operatorname{Hom}_A(R\operatorname{Hom}_A(N, \omega_A^\bullet), E)$$

is an equivalence as in Lemma lemma-equivalence-finite-length. Hence it is isomorphic to the identity functor. Since $\operatorname{Hom}_A(-, E)$ applied twice is the identity (Proposition proposition-matlis) we obtain the statement of the lemma. □